

Correction oscillateur amorti

1. A l'équilibre $\sum_i \vec{F}_i = \vec{0}$

$$\Rightarrow \vec{P} + \vec{F}_r = \vec{0} \Rightarrow (mg - k(\ell_{eq} - \ell_0)) \vec{U}_z = \vec{0}$$

$$\Rightarrow \left[\ell_{eq} = \ell_0 + \frac{mg}{k} \right]$$

2. • $\sum_i \vec{F}_i = m\vec{a} = \vec{P} + \vec{F}_r + \vec{f}$

$$(mg - k(\ell - \ell_0) - \lambda \dot{x}) \vec{U}_x = m a \cdot \vec{U}_x$$

$$\Rightarrow (mg + k\ell_0 - k\ell - \lambda \dot{x}) \vec{U}_x = m \ddot{x} \vec{U}_x$$

$$\Rightarrow +k\ell_{eq} - k\ell - \lambda \dot{x} = m \ddot{x}$$

on projette
selon $\vec{U}_x$

$$\Rightarrow -kx - \lambda \dot{x} = m \ddot{x}$$

$$\Rightarrow \ddot{x} + \frac{\lambda}{m} \dot{x} + \frac{k}{m} x = 0$$

$$\left[\ddot{x} + \frac{\omega_0}{Q} \dot{x} + \omega_0^2 x = 0 \right]$$

• avec $\left[\omega_0 = \sqrt{\frac{k}{m}} \right]$ et $\left[Q = \frac{1}{\lambda} \sqrt{k m} \right]$

3. Régime oscillant si $Q > \frac{1}{2}$

$$\Rightarrow \frac{1}{\chi} \sqrt{Rm} > \frac{1}{2} \Leftrightarrow 2\sqrt{Rm} > \chi$$

4. $\begin{cases} x(0) = 0 \\ \dot{x}(0) = v_0 \end{cases} \Rightarrow [A = 0]$

$$x(t) = -\frac{1}{\tau} e^{-t/\tau} (A \cos(\Omega t) + B \sin(\Omega t)) \\ + e^{-t/\tau} (-A \Omega \sin(\Omega t) + B \Omega \cos(\Omega t))$$

$$\dot{x}(0) = -\frac{A}{\tau} + B \Omega = B \Omega = v_0$$

$$\Rightarrow [B = \frac{v_0}{\Omega}]$$

$$[x(t) = \frac{v_0}{\Omega} e^{-t/\tau} \sin(\Omega t)]$$

5. $[T = \frac{2\pi}{\Omega} = \frac{2\pi}{\omega_0 \sqrt{1 - \frac{1}{4Q^2}}}]$

6. $\delta = \ln \left(\frac{x(t)}{x(t+T)} \right), \quad \frac{x(t)}{x(t+T)} = \frac{1}{e^{-T/\tau}} = e^{T/\tau}$

$$\delta = \frac{T}{\tau} = \frac{2\pi}{\frac{2Q\omega_0 \sqrt{1 - \frac{1}{4Q^2}}}{\omega_0}} = \left[\frac{\pi}{Q \sqrt{1 - \frac{1}{4Q^2}}} = \delta \right]$$

7. $T = t_c - t_0 = 1,1 \text{ s}$ ou $\frac{t_0 - t_1}{7}$
 $\delta = \ln \left(\frac{x(t_0)}{x(t_c)} \right) = 0,109 \approx 0,11$

$$\Rightarrow Q = ?$$

$$Q^2 \delta^2 \left(1 - \frac{1}{4Q^2} \right) = \pi^2 \Rightarrow Q^2 - \frac{1}{4} = \left(\frac{\pi}{\delta} \right)^2$$

$$\Rightarrow Q = \sqrt{\left(\frac{\pi}{\delta} \right)^2 + \frac{1}{4}} = 28,56$$